Welcome To The Bitcoin Diploma

About Us

My First Bitcoin is a nonprofit education organization founded by John Dennehy in El Salvador in 2021. We publish free and open-source Bitcoin education for learners and educators around the world.

The project began with small in-person classes in El Salvador in 2021. The 90-minute introduction moved from what money is to how Bitcoin works, then gave the learner practical experience sending and receiving sats with a Lightning wallet. Within two months, the project was teaching hundreds of learners each month. My First Bitcoin's public history records the project's milestones and organizational accountability.

In 2022, the first Bitcoin Diploma cohort began in one public school in San Marcos, El Salvador. The pilot developed into an open-source curriculum that independent educators could translate, adapt, and teach. It later spread through the My First Bitcoin Education Network to communities around the world.

Our work is independent, community-driven, and Bitcoin-focussed. We focus on financial literacy, careful thinking, and practical safety—not trading, speculation, or promoting other tokens. We do not ask you to accept a belief. Instead, we provide concepts, questions, and practical knowledge so you can form your own conclusions.

Our mission: independent, open-source Bitcoin education

Our mission is to empower people through independent, impartial, and open-source Bitcoin education.

Independent education is not controlled by a government, company, or financial service. Impartial education explains evidence and trade-offs without requiring a learner to reach a predetermined conclusion. Open-source education can be read, shared, reviewed, translated, and improved by anyone under its license.

This approach matters because money affects every person. People should be able to study the systems they use without needing permission. They should also be able to inspect the educational material, question its claims, and help improve it.

What we stand for: accessibility, critical thinking, community, and self-sovereignty

Accessibility means that Diploma content can be taught in person (preferred), online, or through ~~low-tech~~ print/PDF delivery. Limited technology should not prevent participation in learning or certification. Local examples may change, but the required competencies and competency criteria remain the same.

Curiosity means being willing to ask why, admit what you do not know, and investigate an unfamiliar answer before accepting or rejecting it.

Critical thinking means asking clear questions, checking evidence, and comparing trade-offs. Bitcoin makes strong claims about money and ownership. These claims deserve careful study. During this course, distinguish facts from opinions, and possibilities from guarantees.

Community means that education becomes more useful when people discuss ideas and connect them to local life. Your educator and classmates may have experiences that no global book can contain. Listen carefully, ask questions, and explain your reasoning with respect.

Self-sovereignty means having meaningful control over your own choices and resources. With Bitcoin, greater control can reduce dependence on institutions. It also creates greater personal responsibility. Control without knowledge can be dangerous, so practical safety is a central part of this Diploma.

Learn more with the My First Bitcoin ~~App~~

The My First Bitcoin ~~app~~ is an optional resource. It can offer short self-check questions, recap material, and ways to discover more from My First Bitcoin. You are also able to use it to find ~~a My First Bitcoin~~ educator ~~or Diploma learning opportunity~~ near you. Availability depends on your location. The app is not required to take part in the Diploma, complete the learning, ~~or receive a certificate~~. Your book, educator, and ~~group~~ activities provide the complete learning path.

Why ~~this~~ the Bitcoin Diploma exists

Most people use money before they learn how money works. They may earn it, save it, borrow it, and spend it without studying who issues it, how its supply changes, or why its purchasing power changes.

Bitcoin introduces a different system. It has no central issuer, it follows public rules, and it allows direct digital ownership. These ideas can be difficult to

explain clearly without first examining money and the problems that can arise in modern money systems.

This Diploma therefore follows a simple path:

1. Explain the problem Bitcoin was designed to address.
2. Explain Bitcoin's main response.
3. Explain and practise the responsibility of holding and using bitcoin.
4. Explore how Bitcoin may function as money in your community.

The aim is not to make you an expert in eight hours. The aim is to give you a reliable foundation, help you avoid common beginner mistakes, and prepare you to continue learning.

How learners can use this course as a starting point

Treat this Diploma as the beginning of a longer learning process. By the end, you should know enough to ask better questions and to recognize which topics require more study.

You can use the course to:

- explain common terms about money and Bitcoin;
- compare modern money systems and Bitcoin in plain language;
- discuss claims using evidence and trade-offs;
- identify wallet and custody choices;
- recognize common safety risks;
- choose a safe next learning step;
- connect with a local learning community;
- prepare for more advanced study.

You do not need to buy bitcoin to complete this Diploma. You also do not need to install a Bitcoin wallet or use personal funds. Required practical skills may be assessed with a prepared wallet, simulation, testnet, or an educator-approved low-stakes environment. Knowledge comes before action.

If you have limited time or would like a first touch point before getting to grips with Bitcoin, begin with Introduction to Bitcoin (ITB). It is a shorter course that introduces the basics. After the Diploma, learners who want to teach can continue to the Educator Training Program (ETP). These are separate learning paths: ITB is a short primer, the Diploma builds a broader beginner foundation, and ETP develops teaching practice.

How This Bitcoin Diploma Works

What learners will be able to explain and demonstrate by the end

After completing this Diploma, you should be able to:

- explain what money is and why people use it;
- identify the functions and useful properties of money;
- describe the path from barter to modern money;
- explain inflation, purchasing power, and fractional reserve banking, ~~through essential concepts~~;
- explain why Bitcoin was created and how its rules differ from those of modern money systems;
- describe decentralization, finite supply, peer-to-peer transfer, and verification;
- identify the roles of users, nodes, and miners;
- explain wallets, addresses, private keys, seed phrases, and custody;
- compare custodial and non-custodial use, as well as hot and cold wallets;
- safely demonstrate required beginner wallet, receiving or sending, recovery-readiness, and transaction-verification actions ~~under supported conditions~~;
- distinguish on-chain and Lightning payments;
- recognize common scams and basic safety practices;
- identify different ways people may acquire or exchange bitcoin without treating one path as a recommendation;
- discuss how Bitcoin may function as money in a local community.

Very good.

These competencies are a foundation, not proof of expert skill. Completing the Diploma does not mean you are ready to manage large amounts of bitcoin, advise other people, or solve advanced technical problems without support.

Flexible learning: ~~in person, online, or through low-tech delivery~~

My First Bitcoin prefers in-person Diploma sessions because direct contact with an educator and more opportunities for practical activities can make learning more engaging and natural. This is a preference, not a requirement. Online teaching can be more affordable, scalable, and achievable for many people, so the Diploma supports both modes.

Diploma content may be taught in person, online, through a blended course, or with ~~low-tech~~ print/PDF materials. The book provides the shared knowledge path. Your educator adds discussion, guided activities, demonstrations, local context, and any required assessment.

Each module follows the same rhythm:

1. Learn a key idea.
2. Show what you can explain or apply.
3. Apply the idea to a comparison, question, scenario, or demonstration.
4. Explain the idea in your own words.
5. Review how it connects to the next module.

You will learn more by participating than by reading silently. Some questions do not have one correct answer. What matters is whether you can explain your reasoning, examine evidence, and revise your view when you learn something new.

Participation expectations

Come prepared to think, ask, listen, and contribute. You are not expected to agree with every claim or with every person in your cohort.

During discussion:

- describe ideas in your own words;
- ask for clarification when a term is unclear;
- separate personal experience from general evidence;
- challenge ideas without attacking people;
- state uncertainty when you are unsure;
- remember that financial access, security, and opportunity differ; do not assume that everyone shares your circumstances;
- protect other people's private financial information;
- treat personal stories shared in the cohort as private unless the person gives permission to repeat them;
- do not photograph, record, or share another person's screen, work, or identifying information without permission;
- do not pressure anyone to buy, sell, or reveal how much money they hold.

Your educator will explain any required assessment activities. Theory is mainly developed in Modules 1 and 2, where your educator observes you explain, compare, justify, or apply an idea. A statement of confidence is not evidence by

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itself. Practical competency is mainly developed in Modules 3 and 4, where an educator observes the relevant actions under safe, supported conditions.

You can view the assessment grid your educator uses.

Remark

To receive a certificate, you must meet the approved Diploma requirements. Attendance, app use, or completing a self-check quiz alone is not enough.

Use the My First Bitcoin app if it helps you

Remark
It's a duplicate

You may use the app for short self-checks, recap, or to find more My First Bitcoin learning. Its quiz is a learning aid, not an assessment or a Bitcoin wallet. If you do not have a suitable device or connection, you can still participate fully in the Diploma.

Required prior knowledge: none

No prior knowledge is required. The course introduces each required term before using it in greater depth.

Basic reading and everyday arithmetic are useful. The most important calculation is that one bitcoin can be divided into 100,000,000 smaller units called satoshis. A calculator may help, but advanced mathematics is not required.

If you already use Bitcoin, support your educator, share your experience, and explain ideas to peers in plain language. Teaching others is one way to consolidate your own learning. If you enjoy supporting learners, contribute feedback or local community support, and explore the Educator Training Program (ETP) to develop teaching practice.

Required materials and technology expectations

For the Diploma, you need:

- this book in print or PDF form;
- a notebook and pen, or another way to record answers;
- an educator or study group where available;
- optional access to the My First Bitcoin app if you would like to use its self-check and recap resources;
- time to discuss and reflect.

A personal phone and continuous internet connection are not required for learning, practical preparation, assessment, or certification. Do not install a

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price. It does not recommend that you buy, sell, borrow against, or hold any amount of bitcoin.

The Diploma is Bitcoin-only. Other tokens, trading products, and broader cryptocurrency promotion are outside its scope. This boundary defines the subject of the course; it is not a guarantee about Bitcoin or a recommendation to acquire it.

Bitcoin's market price can change quickly. Transactions can be difficult or impossible to reverse. A lost seed phrase, exposed private key, dishonest custodian, software error, or scam can cause permanent loss.

Use three rules throughout the course:

1. **Learn first.** Do not act only because of excitement, fear, or pressure.
2. **Start small.** Test unfamiliar tools and processes with an amount you can afford to lose.
3. **Don't Trust, Verify.** Check addresses, amounts, fees, sources, and people before you approve an action.

Responsibility also includes knowing when to ask for help. Seek help before moving meaningful funds, but never reveal a seed phrase or private key to the person helping you.

The four-module Diploma, ~~educator-curated Module 4, and Advanced learning~~

The Diploma consists of a four-stage learning arc: **Problem → Solution → Action → Connect.** All four modules are part of the Diploma. Modules 1 and 2 develop the shared Problem and Solution through explanation, comparison, and reasoning. Educators can adapt practical examples, tools, resources, and delivery in Modules 3 and 4 while preserving the required safety and competency standards. Module 4 is the educator-curated Connect stage, where learners link the shared learning to a local need or opportunity, a realistic alternative, safe practice, and a responsible next step.

Module 4 is a required part of the Diploma, curated by the educator. Educators are the subject-matter experts on their students' local constraints, solutions, tools, resources, laws, language, infrastructure, and needs. The educator-curated Connect experience may include earning, saving, receiving, spending, Layer 2 solutions, local exchange pathways, merchants, or circular economies.

Advanced learning is outside the Diploma. It may examine deeper economics, cryptography, mining, privacy, transaction structure, Lightning operations,

business use, or security. Advanced material is kept separate so the ~~beginner~~ Bitcoin Diploma remains clear and safe.

A topic is not unimportant because it is outside the Diploma. It may simply require more time or prior knowledge.

How the ~~4 module~~ learning path is organized

#	Module	Role	Guiding question	Learner promise
1	~~Module 1:~~ Understanding Money	Problem	How can money break? ↑ Reformulate	You can explain the money problems Bitcoin responds to.
2	~~Module 2~~: How Bitcoin Works	Solution	Why was Bitcoin created, and how does it respond?	You can explain Bitcoin's purpose and key properties.
3	~~Module 3~~: Using Bitcoin Safely	Action	How do I use and secure Bitcoin responsibly?	You can explain beginner custody and safety concepts and act safely in supported practice.
4	~~Module 4~~: Bitcoin in Your Community	Connect	How do the problem, solution, and safe action connect to this community?	You can connect the course learning to a local need, alternative, safe task, and responsible next step.

The four modules provide the required Problem → Solution → Action → Connect learning path, not a fixed class timetable. Modules 1 and 2 develop theory-based competencies through explanation, comparison, and reasoning. Module 3 develops safe practical action. Module 4 is educator-curated but required: it connects that shared learning to a local need, realistic alternative, safe task, and responsible next step. Four two-hour sessions are one possible format. Your educator may divide a module across more sessions, revisit an earlier idea, or change the order of individual sections when your cohort needs more time for discussion, practice, review, or assessment. The educator remains responsible for covering all required ~~Diploma~~ competencies and making the connections among the modules clear.

My engagement promise

I, _____, intend to take part with curiosity, care, and respect. I will ask questions, think for myself, and use this learning responsibly.

My intention for this Diploma:

_____

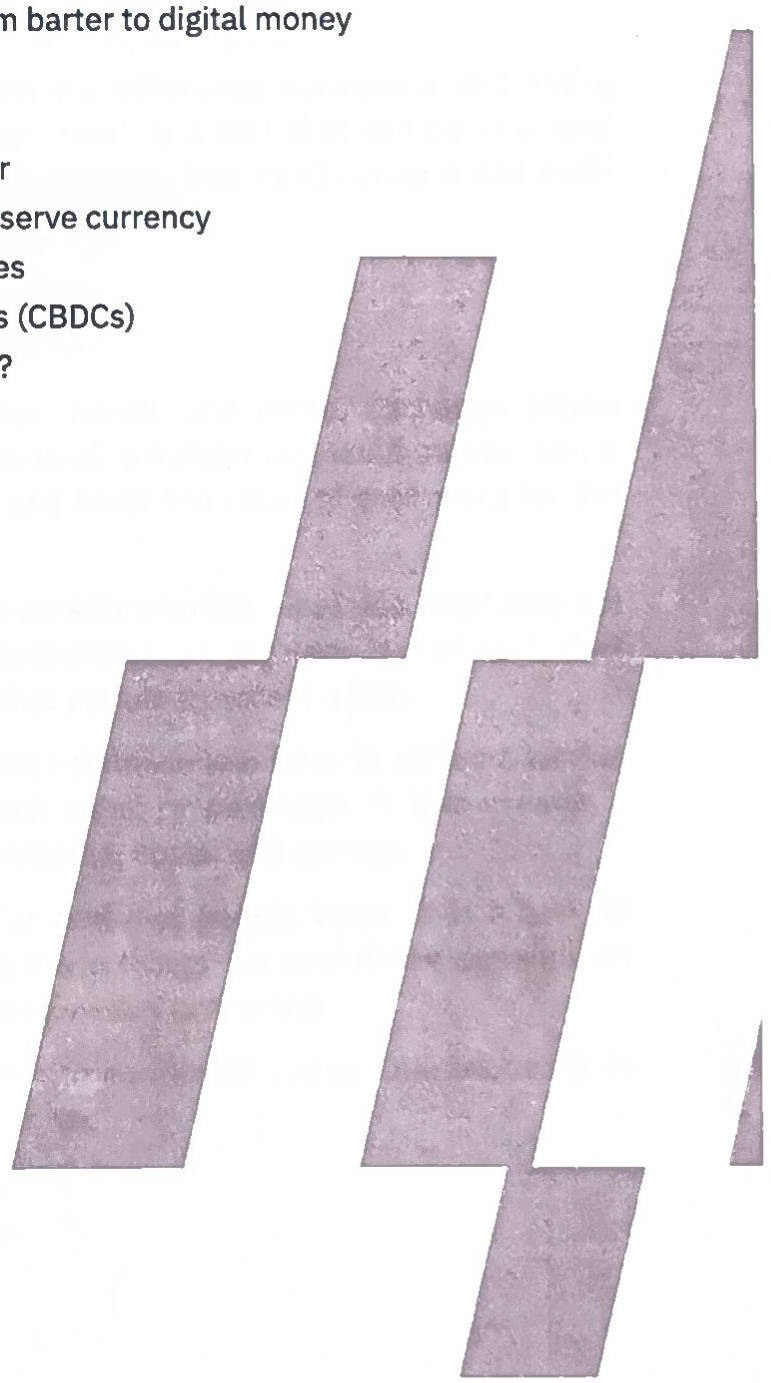
Signature: _____ **Date:** _____

Remote

Module 01

Understanding Money

1.1 Opening questions about money
1.2 Why money exists
1.3 Functions and properties of money
1.4 Barter and the double coincidence of wants
1.5 Scarcity, value, and exchange
1.6 Beginner monetary history from barter to digital money
1.7 Fiat money
1.8 Fractional reserve banking
1.9 Inflation and purchasing power
1.10 Why the US dollar is a world reserve currency
1.11 Why people look for alternatives
1.12 Central Bank Digital Currencies (CBDCs)
1.13 Review: how can money break?



1.1 Opening questions about money

~~**Learner promise:** You will explain the money problems Bitcoin responds to.~~

Opening reflection | Suggested time: 10 minutes | Individual writing, then cohort discussion

avoid "not simple"

reformat

Money is familiar, but it is not simple. Before the course discussion begins, write a short answer to each question.

1. Why do people need money?
2. What makes something money?
3. Who can create or control the money you use?
4. Why does the money you use have value?
5. Does your money buy the same amount today as it did in the past?

Keep your answers. Return to them at the end of the Diploma. This course may strengthen some of your views and change others.

As you discuss these questions, notice the difference between a fact and a preference. "This note is accepted for taxes" is a fact that can be checked. "This is the best form of money" is a conclusion that needs criteria and evidence.

1.2 Why money exists

People have different skills, resources, needs, and plans. Exchange allows them to specialize. A farmer can grow food, a builder can build homes, and a nurse can provide care. Each person can trade the value of their work for the work of others.

Direct exchange is often difficult. The person who has what you need may not want what you offer. Money solves this coordination problem. It is an asset that people accept because they expect other people to accept it later.

Money helps strangers cooperate across distance and time. It allows a worker to receive value today and use it next week or next year. It also creates a common way to record prices, debts, income, costs, and savings.

Money is not the food, home, time, or skill that people value. It is a tool for measuring and exchanging claims on those things. Its usefulness depends on trust, rules, social acceptance, and its monetary properties.

Spot question: Why did money come into existence? Explain the exchange or coordination problem it helps people solve.

1.3 Functions and properties of money

Money usually performs three main functions.

1. **Medium of exchange:** People accept it in payment for goods and services. This reduces the need for direct barter.
2. **Unit of account:** Prices are expressed in a shared unit. This allows people to compare the cost of different goods and keep financial records.
3. **Store of value:** Value can be moved from the present into the future. No money stores value perfectly, but some forms lose purchasing power more slowly than others.

An item can perform one function better than another. A national currency may be widely accepted for daily payments while another asset may be chosen for long-term saving.

Useful money also tends to have several properties:

Property	Question to ask
Durability	Can it remain usable over time?
Portability	Can people move or transfer it without excessive difficulty?
Divisibility	Can it represent both small and large amounts?
Fungibility	Are equal units normally interchangeable?
Scarcity	Is its supply difficult to increase without limit?
Acceptability	Will other people receive it in exchange?
Verifiability	Can people check that it is genuine?

No form of money scores perfectly. Gold is durable and scarce but difficult to divide and transport for small payments. Paper cash is portable and accepted locally, but it can be damaged and its supply can expand. The best comparison depends on the purpose, place, and time.

Think critically: Name a time when you chose one type of money or asset over another. What was the purpose? Compare the properties of the options you could have used. Which properties mattered most to you, and why?

1.4 Barter and the double coincidence of wants

Barter is the direct exchange of one good or service for another. It can work well in a small exchange. It becomes difficult when many people and products are involved.

Imagine that Amina has rice and wants shoes. Luis has shoes but wants cooking oil, not rice. Mei has cooking oil and wants repair work. Amina may need several exchanges before she can obtain the shoes. Each exchange takes time and requires agreement about value.

For barter to work directly, each person must want what the other person offers at the same time and in suitable amounts. Economists call this the **double coincidence of wants.**

Barter also creates other problems:

- some goods cannot be divided without destroying their value
- some goods spoil before they can be exchanged
- there is no common unit for comparing prices
- saving large amounts can be difficult
- delayed payment and accounting become complex

A commonly accepted money separates selling from buying. Amina can sell rice to one person for money and later use that money to buy shoes from someone else.

1.5 Scarcity, value, and exchange

Resources are limited, while human wants are not. **Scarcity** means that choosing one use of a resource prevents another use. Time, land, energy, good health, tools, and money are all scarce.

The value of a good is not contained inside the object. People value goods according to their needs, preferences, information, and circumstances. Clean water may be inexpensive near a safe public source and extremely valuable during a shortage.

Exchange happens when both sides prefer what they receive to what they give up. They do not need to agree that the two items have the same personal value. They only need to agree to the terms.

Three ideas help explain everyday choices:

- **Opportunity cost** is the best alternative you give up when you make a choice. Money spent on transport cannot also be spent on food.
- **Time preference** describes how strongly a person prefers value now instead of later. Urgent needs can create a high preference for the present. Saving for a future goal shows a greater willingness to wait.
- **Trade-off** means gaining one benefit while accepting a cost or giving up another benefit.

Scarcity alone does not guarantee value. An item may be rare and still have no demand. A useful monetary asset combines limited supply with properties and demand that make people willing to hold or exchange it.

1.6 Beginner monetary history from barter to digital money

Money did not develop in one identical sequence everywhere. Different societies used different goods, records, and institutions. However, a broad pattern helps explain modern money.

> **"Money ... is by no means the result of an agreement ... but is the unintended product of historical development."**
>
> **— Carl Menger, "On the Origin of Money" (1892)**

Direct exchange and local records: People used barter, gifts, obligations, and written or remembered debts. These systems depended strongly on local relationships.

Commodity money: Communities selected goods that were widely valued and easier to trade. Salt, beads, shells, livestock, copper, silver, and gold have served monetary roles in different places. A commodity has a use or value beyond an issuer's promise.

Standard coins: Authorities measured and stamped metal into standard units. Coins improved recognition and accounting. Authorities could also reduce the precious-metal content of these coins while keeping the same stated value. This is called debasement.

Representative paper: Transporting and protecting metal was difficult. Depositories and banks issued paper claims that could be redeemed for a stated

amount of metal. The paper was easier to move, but users had to trust the issuer to keep enough metal and honor redemption. Under different versions of a **gold standard**, a currency unit was legally connected to a stated amount of gold. The rules varied by place and period, and governments or banks sometimes suspended or ended convertibility.

Bank and government money: Banks issued more claims and credit. Governments created central banks and national currency systems. Money became increasingly based on institutional records and promises rather than direct possession of a commodity by the population. In many countries, national currency later became no longer redeemable for a physical asset such as gold. This separation from commodity backing is a defining feature of today's **fiat money** (see below).

Digital money: Much of today's money moves as digital records through banks and payment services. A card or phone does not create a new kind of money by itself. It is usually an interface to a bank deposit or another claim denominated in the national currency.

This history shows repeated trade-offs. Convenience increased, but users often became more dependent on monetary policy, issuers, banks, and payment intermediaries.

1.7 Fiat money

Fiat money is currency issued under government authority, usually through a central bank, that is not redeemable for a fixed amount of a commodity such as gold. The word fiat is Latin for "let it be done," often understood as "by decree." A country's laws typically designate its currency as legal tender and require it for the payment of taxes, although the exact rules for accepting payments and settling debts vary by jurisdiction.

Fiat money works because people, businesses, banks, and governments accept it. Network effects reinforce this acceptance: people want the currency because other people use it and prices are stated in it.

The fiat system has several main participants:

Participant	Main role
Central bank	Issues base money, manages monetary policy, supports payment settlement, and may act as lender of last resort.
Commercial bank	Holds customer deposits, provides payments, and creates deposit money when it issues loans.
Government	Collects taxes, spends, borrows, creates laws, and usually defines the national unit of account.
Citizens and businesses	Earn, save, borrow, invest, pay taxes, set prices, and exchange goods and services.

Local scenario: Choose one ordinary payment in your community, such as buying transport, food, or phone credit; receiving wages; or paying a bill. Use a realistic invented scenario if you prefer. In three or four sentences, explain:

1. what is being paid for and which currency is used as the unit of account;
2. why the seller or recipient is willing to accept that currency; and
3. which institutions, records, or rules people rely on to complete the payment.

Finish by naming one way the payment would become harder if that shared acceptance or trust broke down.

Fiat systems can provide convenient payments, flexible credit, and tools for responding to financial crises. The same flexibility creates **risk**. A small number of institutions can change the supply and cost of money, restrict access, or make decisions that reduce purchasing power.

Fiat money requires layers of **trust**. Users trust banks to keep accurate records, payment services to process transfers, central banks to manage the currency, and governments to maintain lawful institutions. These systems can work for long periods, but their rules are not fixed and can be open to neglect or abuse.

1.8 Fractional reserve banking

When you deposit cash in a bank, the bank records a liability to you: it owes you the account balance. The bank does not normally keep every unit of every deposit as physical cash. It holds liquid reserves and uses its balance sheet to make loans and other investments.

When a commercial bank approves a loan, it usually creates a new deposit in the borrower's account. This expands the amount of bank deposit money. When the principal is repaid, that deposit money is reduced. Banks are limited by capital rules, liquidity needs, credit risk, regulation, the cost of funding, and demand from qualified borrowers.

Banks generally seek profit by earning interest and fees on loans and investments while paying their operating, funding, and regulatory costs and absorbing losses when borrowers do not repay. These activities and constraints differ by bank and jurisdiction.

Fractional reserve banking is the broad idea that banks keep only a fraction of their deposit obligations available as immediately accessible reserves. Exact rules differ between countries and can change over time.

This creates two important risks:

- **Liquidity risk:** A bank may own long-term loans that are expected to be repaid but still lack enough cash for many withdrawals today.
- **Solvency risk:** Failed loans or falling asset values may leave the bank's total assets worth less than what it owes.

A **bank run** happens when many depositors try to withdraw at the same time because they fear the bank may fail. Deposit insurance and central-bank lending can reduce panic, but insurance eligibility and coverage limits vary by jurisdiction. These protections depend on institutions and rules, and they can move some losses or risks from an individual bank to the wider system.

Regulation and oversight can reduce risk, but they cannot prevent every case of fraud, poor management, political abuse, or simple error. These failures are possibilities to evaluate, not outcomes of every banking system.

The central lesson is not that every bank is about to fail. It is that a bank balance is the bank's promise to pay you. It is not the same as holding the settlement asset directly.

1.9 Inflation and purchasing power

Price inflation is a sustained increase in the general price level of goods and services. If your income and savings do not rise as quickly as prices, your money buys less.

Purchasing power is the amount of goods and services a unit of money can buy. If a basket of basic goods rises from 100 units of currency to 110, the same 100 units no longer buy the full basket. Its purchasing power in relation to that basket has fallen.

Inflation can have several causes. These include an expansion of money and credit, supply disruptions, changes in demand, currency weakness, taxes, regulation, and changes in production costs. The causes and effects are often debated. Avoid explanations that assign every price change to one factor.

Inflation does not affect everyone equally. A person who holds mainly cash may lose purchasing power. A person who owns an asset that rises in price may be partly protected. Borrowers may repay fixed debts with money that buys less, while lenders may receive less real value than expected. Wages and prices also adjust at different speeds.

The rate of inflation is an average. Your personal experience may differ because your spending is not identical to the measured basket. Food, housing, transport, education, and health costs may change at different rates.

Purchasing-power check: Choose one ordinary item in your community. Record its price today. Ask what it cost several years ago, then check a reliable record if possible. Did income change at the same rate? What other factors may explain the change?

Then compare your item with an official inflation measure. Which goods and services are in that measure's basket? Has the basket or method changed over time, and does the source explain why? A change is not automatically manipulation, but it can affect comparisons and should be documented transparently.

1.10 Why the US dollar is a world reserve currency

A **reserve currency** is a currency that central banks and governments hold and use for international payments. Before the First World War, the British pound played this role most strongly. The war weakened European currencies and financial centres, while the United States supplied goods and accumulated gold. This increased international use of the dollar. After the Second World War, the United States had a large economy and much of its financial capacity intact. The 1944 Bretton Woods Agreement then organized a postwar system in which many currencies were linked to the dollar and the dollar was officially linked to gold, placing the dollar at the centre of international payments.

In 1971, President Nixon ended the official exchange of dollars for gold by foreign governments and central banks, helping bring the Bretton Woods system to an end. The dollar nevertheless remained widely used because of its established role in trade and finance, the size of the U.S. economy, and the depth of U.S. financial markets. It remains the leading reserve currency today, al-

though central banks also hold other currencies and its share can change over time.

1.11 Why people look for alternatives

People often seek alternatives when they believe cash will lose purchasing power. Common choices include precious metals, real estate, and stocks. Each has benefits and risks.

Alternative	Why people may choose it	Important trade-offs
Precious metals	Long history, physical possession, limited supply	Storage, verification, transport, security, and difficult small payments
Real estate	Useful physical asset, possible rent, limited desirable locations	High entry cost, maintenance, taxes, legal dependence, and low portability
Stocks	Ownership in productive businesses, possible income and growth	Business risk, market volatility, custody through institutions, and need for financial knowledge

These assets are not identical to money. A home may store value but is not practical for buying food. A share may be easy to sell in one country but unavailable to someone without a brokerage account. Gold may cross long periods of history but be difficult to verify or divide during a small exchange.

People may also choose foreign currency, government bonds, private businesses, land, tools, education, or other assets. Every choice includes opportunity cost and risk. There is no asset that guarantees purchasing power in every period and situation.

Note: Tax treatment can also shape these decisions. Rules for income, gains, property, and transactions can change the cost or benefit of holding, selling, or using an asset. Tax rules vary by jurisdiction and can change over time; this course does not provide tax advice.

Bitcoin emerged as another proposed alternative: a scarce digital asset that can be held and transferred without a central issuer. To evaluate that claim, we must study why Bitcoin was created and how its rules work.

1.12 Central Bank Digital Currencies (CBDCs)

A **central bank digital currency**, or **CBDC**, is a digital form of fiat money issued as a liability of a central bank. It is different from physical cash and from a normal commercial-bank deposit, although the user experience may look similar.

CBDC designs differ. Some are intended mainly for financial institutions. Others are proposed for public use. Depending on the design, a CBDC may allow faster settlement, wider access, or new payment features. It may also give institutions greater ability to monitor transactions, apply rules, limit use, or connect identity directly to payments.

A CBDC does not create a fixed monetary supply or remove central control. It changes the technology and distribution of fiat money. The central bank remains the issuer and can still apply monetary policy.

Digital does not automatically mean decentralized, scarce, private, or open to independent verification. A CBDC may use digital technology, but it is still fiat money if its issuer, supply rules, access rules, and payment permissions remain centrally managed.

When discussing a CBDC, ask:

- Who issues the money?
- Who can hold or use it?
- What information is collected?
- Can a payment be blocked, limited, or reversed?
- Can the rules change?
- Can ordinary users independently verify the rules?
- What happens without electricity or internet access? -> same for bitcoin
- What legal protections exist?

Do not assume that every proposed CBDC has the same design. Use current, local, and reliable information.

1.13 Review: how can money break?

Pause and consider the money you use: what makes it useful in daily life, and which of its rules depend on institutions you must trust? Barter is difficult because matching needs is inefficient, while fiat money makes exchange easier but can involve changing supply, restricted access, and declining purchasing power. In short, people look for monetary alternatives when they want different trade-offs around supply, control, access, and trust.

Key takeaways

- Money helps people exchange, compare prices, record value, and move value across time.
- The main functions of money are a medium of exchange, a unit of account, and a store of value.
- Monetary properties such as durability, portability, divisibility, scarcity, acceptability, and verifiability help us compare different forms of money.
- Barter is difficult at scale because both people must want what the other offers at the same time and in suitable amounts.
- Money has developed in different ways across societies, often improving convenience while creating new trade-offs in trust, control, and access.
- Fiat money can support convenient payments and flexible credit, but its supply and rules depend on institutions.
- Inflation, fractional-reserve banking, and CBDC design can affect purchasing power, access, privacy, and the need to trust institutions.
- Some people look to alternatives such as precious metals when they want different trade-offs around purchasing power, supply, or control; every alternative has limits and risks.

The next module examines Bitcoin's attempt to create a different set of monetary rules.

Module 02

How Bitcoin Works

2.1 Why Bitcoin was created
2.2 The search for digital and decentralized money
2.3 Early digital-money attempts and the double-spend problem
2.4 Digital scarcity and credible scarcity
2.5 The Blockchain - Bitcoin's Public Ledger
2.6 Decentralization
2.7 Digital signatures
2.8 Peer-to-peer transfer
2.9 Users, nodes, and miners
2.10 Proof of work
2.11 Finite supply
2.12 Satoshis and unit bias
2.13 Bitcoin as sound digital money
2.14 Bitcoin, stablecoins, and other crypto-assets
2.15 Learning and responsible use are not trading or promotion
2.16 Bitcoin origin timeline
2.17 Participant empowerment, risk, and responsibility
2.18 Review: how does Bitcoin respond to fiat problems?

2.1 Why Bitcoin was created

Learner promise: You will explain why Bitcoin was created and what makes it different.

Digital payments existed before Bitcoin. People could send money through banks, card networks, and online payment companies. These systems made commerce easier, but each depended on a trusted organization to keep the ledger, approve accounts, settle transfers, and resolve disputes.

This dependence creates power and risk. An intermediary can delay a payment, charge a fee, collect and misuse personal data, restrict an account, reverse a transfer, or fail. The issuer of the currency can also change its supply.

For many years, computer scientists tried to create digital money that did not need a central issuer or payment company. The challenge was difficult. Digital information is easy to copy. If a digital coin could be copied and spent twice, it could not function as money. If one company prevented copying, the company became the point of control and failure.

Bitcoin was created to solve this problem with a peer-to-peer network. Its design allows participants to agree on one transaction history without a central ledger keeper. It combines a public ledger (blockchain), digital signatures, a network of validating nodes, proof of work, and economic incentives.

Bitcoin does not remove trust from every part of life. It reduces the need to trust a central authority for the issuance and settlement of its native money.

In this book, **Bitcoin** with a capital B usually means the network, protocol, or wider system. **bitcoin** with a lowercase b means the monetary unit or asset. A wallet or exchange is a service that may connect to Bitcoin; it is not the Bitcoin network itself.

2.2 The search for digital and decentralized money

The growth of personal computers and the internet created new opportunities for communication. It also created new forms of surveillance and control. In the late twentieth century, a group of programmers, mathematicians, and activists known as **Cypherpunks** explored how cryptography could protect privacy and individual freedom in digital systems.

Cryptography is the use of mathematical methods to protect information and prove that an action is authentic. Encryption can keep a message private. **Digit-**

al signatures can prove that a message or instruction came from the holder of a secret key without revealing that key.

The Cypherpunks understood that private communication was not enough. People also needed a way to exchange value online. If every digital payment required a bank or company, that intermediary could observe, permit, or prevent the exchange.

Their goal was not simply digital money. Bank deposits were already digital. The harder goal was **decentralized digital money**: a system with no single issuer, server, or administrator that participants had to trust.

Bitcoin was built upon decades of work by many people. Its design joined earlier ideas in a new way and provided incentives for an open network to keep operating.

2.3 Early digital-money attempts and the double-spend problem

Bitcoin was not the first attempt at digital money. It was built on four decades of experimentation, drawing on ideas from many earlier projects that helped shape its successful design.

DigiCash used cryptography to support private digital payments. It depended on a central company and banking relationships. When the company failed, the system could not continue as an independent money network.

Hashcash used proof of work to make unwanted email costly. A sender's computer had to perform work before sending a message. The idea showed how computational cost could discourage abuse.

Bit Gold described a proposed system in which computational work could create scarce digital records. It was not launched as a complete decentralized money system, but it developed important ideas about digital scarcity and proof.

B-money and **reusable proof of work** explored ways for participants to create or transfer scarce digital value without relying only on one central database. They were incomplete, but they helped clarify what a working system would need.

BitTorrent was not money. It showed that a peer-to-peer network could let many independent computers share information without one central server controlling the whole system.

Together, these experiments showed useful pieces of the puzzle. They also clarified the remaining problem.

The central problem was **double spending**. A paper note cannot be given to two people at the same time. However, digital data can be copied perfectly. A dishonest user could try to send the same digital unit to two recipients.

Centralized systems prevent this by letting one trusted database decide which payment came first. On 31 October 2008, Satoshi Nakamoto's whitepaper, *Bitcoin: A Peer-to-Peer Electronic Cash System*, explained a different approach: a public system that could establish a shared payment history without a bank, company, or other central authority.

The whitepaper brought together the features needed for that approach:

1. **A peer-to-peer network:** transactions are announced directly to a network of participants.
2. **Digital signatures:** only the holder of the relevant private key can authorize a spend.
3. **Blocks and mining:** miners collect transactions into blocks and use proof of work to compete to add the next block.
4. **Independent verification:** nodes check transactions and blocks against Bitcoin's rules.
5. **A shared history with accumulated proof of work:** participants recognize the valid chain of transactions with the most accumulated work as the shared record.

This module explains these concepts in more detail. Together, they gave a practical answer to the double-spend problem: the network can reject a second attempt to spend the same bitcoin without relying on a central authority. Bitcoin offered the missing combination that decades of digital money experiments had been seeking.

2.4 Digital scarcity and credible scarcity

Digital information is normally easy to copy. Digital scarcity means that the system can distinguish a valid unit from an invalid copy and can prevent the same unit from being spent twice.

Bitcoin creates digital scarcity by combining signatures, a public transaction history, proof of work, and independent rule verification. A bitcoin can be transferred, but the network rejects an attempt to spend the same bitcoin again.

Scarcity is more credible when the rules are public, users can verify them independently, and no single issuer can quietly change the supply for everyone. This is why Bitcoin's supply limit must be studied together with decentralization and node verification.

2.5 The Blockchain - Bitcoin's Public Ledger

Unlike banks, who each have their own ledger of transactions and must coordinate with each other, Bitcoin is recorded and stored in a SINGLE, digital, decentralized, public ledger. This ledger is stored in a data format referred to as a "blockchain".

A block contains a series of transactions (in order), and each block is like a page in a ledger. Transactions within each block are in precise, chronological order and blocks are also sequenced and "chained" together in chronological order.

Bitcoin is designed to add a new block of transactions to the blockchain approximately every 10 minutes. The blockchain is basically append only, and is therefore considered to be "immutable," meaning it cannot be altered, which is a critical concept for accuracy, security, and stability.

2.6 Decentralization

A centralized system has a controlling point. A bank controls its customer ledger. A payment company controls its network. A government controls the issuance rules of its currency.

A **decentralized** system distributes roles and decision enforcement among many independent participants. Bitcoin has no central office that can create accounts, change balances, or approve membership.

Bitcoin's decentralization comes from several features:

- anyone can download and inspect the open-source software;
- users can transact without opening an account with the Bitcoin network;
- independent nodes check transactions and blocks;
- miners compete to propose blocks;
- no miner can make an invalid block acceptable to honest nodes;
- changes to the rules require voluntary adoption by participants;
- the ledger is copied across many computers and locations.

Decentralization is not the same as equal influence. Mining equipment, technical knowledge, money, internet access, and software development are not dis-

tributed equally. Large service providers can become points of dependence for their own users. Bitcoin's claim is narrower: no single participant has the authority to unilaterally change the ledger or monetary rules for everyone.

Decentralization also creates costs. Coordination can be slower. Users must verify information and take more responsibility. There is no central support desk for the Bitcoin network. These trade-offs are part of the design, not accidental defects that can be removed without changing the system.

2.7 Digital signatures

Digital signatures are one of the key parts of Bitcoin's design described in the whitepaper. In Bitcoin, a wallet uses a private key to create a digital signature that authorizes a transaction. A wallet is usually an application that manages the keys and creates these signatures for the user. Only the holder of the required private key can authorize the spend; the network can verify the signature without learning the private key itself.

This allows a person to control bitcoin without asking a bank or other central operator for permission. It also means that control of the private key is critical: anyone who obtains it can usually authorize a spend, while a lost key may make the bitcoin irretrievable.

2.8 Peer-to-peer transfer

Peer-to-peer means that participants interact through a network without requiring a central coordinator. A Bitcoin user can create and sign a transaction, then broadcast it to the network. The recipient does not need an account at the sender's bank because Bitcoin itself provides the common settlement system.

Peer-to-peer does not always mean that only two computers communicate directly. A transaction may pass through wallets, internet providers, and many network nodes. The important point is that no central Bitcoin operator must approve the sender or recipient.

Users can still choose intermediaries. An exchange may hold bitcoin for customers. A payment service may simplify an invoice. A hosted wallet may manage keys. These services can be convenient, but the user then depends on the service's rules and security.

Bitcoin makes direct ownership and transfer possible; it does not force every user to avoid every service. The custody choice determines who has practical control.

Peer-to-peer transfer can be useful when people are in different places or use different banking systems. It also creates responsibility. A valid transaction sent to the wrong address cannot usually be reversed by a central authority.

2.9 Users, nodes, and miners

Bitcoin works because different participants perform different roles.

Users receive, hold, and send bitcoin. A user may control their own keys held in a wallet or use a custodian. Users give the network economic meaning by choosing whether to accept bitcoin and which rules or services they trust.

Nodes run Bitcoin software and independently verify the rules. A full node checks that transactions have valid signatures, that bitcoin have not already been spent, that blocks satisfy proof of work, and that new bitcoin follow the issuance schedule. Nodes share valid data with peers and reject invalid data.

Miners gather pending transactions, build candidate blocks, and compete through proof of work. A successful miner broadcasts a block and may receive the allowed block subsidy and transaction fees. Miners help order transactions and make the ledger costly to rewrite.

Developers review, write, and propose software changes. They do not own Bitcoin's rules. A software change matters only if participants choose to run software that follows those rules.

These roles limit one another:

- a user cannot spend bitcoin without the required valid signature
- a miner can propose a block, but nodes decide whether it follows the rules
- a node verifies using its own rules, but consensus requires compatible rules across the network
- developers can propose software, but participants choose whether to run it

Some people perform several roles. A miner can run a node and also be a user. A beginner does not need to perform every role to explain how the roles balance one another.

Spot question: Why does a node verify transactions? Name at least one rule that a node checks.

2.10 Proof of work

Proof of work is the process miners use to compete for the right to propose the next block.

A miner repeatedly changes data in a candidate block and passes it through a mathematical hash function. The result is unpredictable. The miner is searching for a result below a target set by the network's difficulty rule. There is no shortcut; miners must perform many attempts.

When a miner finds a valid result, it broadcasts the block. Nodes can check the proof quickly, even though finding it required significant computation and energy. If the block includes a transaction, that transaction receives its first **confirmation**. Each valid block added after it adds another confirmation, making the transaction's place in the blockchain more costly to rewrite. This difference is important: work is costly to produce and easy to verify.

Proof of work serves several purposes:

- it gives independent miners a way to compete without a central scheduler;
- it makes the ordering of transactions costly to rewrite;
- it links network security to real-world equipment and energy;
- it supports a public method for choosing between competing valid histories: the chain with the most accumulated work.

Mining does not solve useful equations for an outside purpose. Its purpose is to secure the ordering of Bitcoin transactions. The network adjusts mining difficulty so that blocks continue to arrive about every ten minutes on average as computing power changes.

Proof of work involves a real energy cost. Supporters argue that this cost is what anchors Bitcoin's security and can create demand for flexible or otherwise unused energy. Critics question the amount and source of energy used. A responsible evaluation asks what energy is used, where it comes from, what security it provides, and what alternatives are being compared.

2.11 Finite supply

Bitcoin's rules define a predictable issuance schedule and a maximum supply of 21 million bitcoin. New bitcoin enter circulation through the block subsidy paid to miners who produce valid blocks.

Every 210,000 blocks, the subsidy is cut in half. This event is called a **halving**. Because blocks are produced approximately every ten minutes on average, a halving occurs roughly every four years. The exact calendar date is not fixed.

Halving year	Block height	Subsidy after halving
2012	210,000	25 bitcoin
2016	420,000	12.5 bitcoin
2020	630,000	6.25 bitcoin
2024	840,000	3.125 bitcoin

Add 2028

The subsidy began at 50 bitcoin per block. It falls by half in repeated steps until it becomes too small to continue. Miners also receive transaction fees. Over time, fees are expected to become a more important part of miner revenue.

The 21 million limit is enforced by nodes. If a miner creates more bitcoin than the rules allow, validating nodes reject the block. A software developer can write different code, but cannot force other people to run it.

Bitcoin's supply rule is credible because it is public, easy to verify with a node, and difficult for one party to change. It is not a physical law. It is a network rule that remains effective while participants continue to enforce it.

Many digital assets can claim a limited supply. The beginner question is whether the limit is credible in practice, not only whether it is written in a document or displayed on a website.

Spot question: What are the advantages of Bitcoin having a limited supply? What would be the effect of increasing the supply of Bitcoin on purchasing power? Who would reject an increase in supply?

Finite supply does not guarantee a rising market price. Price depends on demand as well as supply. It also does not mean every bitcoin is available for sale; some may be held long term or become inaccessible because keys were lost.

2.12 Satoshis and unit bias

One bitcoin can be divided into **100,000,000 satoshis**. A satoshi, often shortened to **sat**, is the smallest unit recorded on Bitcoin's base layer.

bitcoin ->

Unit	Amount
1 bitcoin	100,000,000 sats
0.1 bitcoin	10,000,000 sats
0.01 bitcoin	1,000,000 sats
0.001 bitcoin	100,000 sats
0.00000001 bitcoin	1 sat

<- sats

You do not need to buy or use a whole bitcoin. Bitcoin software can display an amount in bitcoin or sats. Many people use sats for everyday amounts because whole numbers are easier to read.

Unit bias is the mistaken feeling that a lower price per whole unit makes an asset cheaper or more accessible. The number of units is a design choice. Owning 100,000 sats and owning 0.001 bitcoin describe the same amount.

When comparing prices, focus on the share of the total supply, the amount of value involved, and the risks. Do not assume that an asset with many cheap-looking units offers better value than an asset divided into smaller units.

-> Compare sats with cents?

Show your understanding: How many sats are in 0.01 bitcoin? Why does the price of one whole bitcoin not prevent a person from using a smaller amount?

2.13 Bitcoin as sound digital money

Sound money is money whose supply is difficult to expand and whose properties support reliable use across time. The term expresses a judgment based on criteria, so it should be examined rather than accepted as a label.

Bitcoin combines properties that did not previously exist together in one asset:

bitcoin -> ← sats

Unit	Amount
1 bitcoin	100,000,000 sats
0.1 bitcoin	10,000,000 sats
0.01 bitcoin	1,000,000 sats
0.001 bitcoin	100,000 sats
0.00000001 bitcoin	1 sat

You do not need to buy or use a whole bitcoin. *Bitcoin software can display an amount in bitcoin or sats. Many people use sats for everyday amounts because whole numbers are easier to read.*

Unit bias *is the mistaken feeling that a lower price per whole unit makes an asset cheaper or more accessible. The number of units is a design choice. Owning 100,000 sats and owning 0.001 bitcoin describe the same amount.*

→ Compare sats with cents?

When comparing prices, focus on the share of the total supply, the amount of value involved, and the risks. Do not assume that an asset with many cheap-looking units offers better value than an asset divided into smaller units.

Show your understanding: *How many sats are in 0.01 bitcoin? Why does the price of one whole bitcoin not prevent a person from using a smaller amount?*

2.13 Bitcoin as sound digital money

Sound money *is money whose supply is difficult to expand and whose properties support reliable use across time. The term expresses a judgment based on criteria, so it should be examined rather than accepted as a label.*

Bitcoin combines properties that did not previously exist together in one asset:

Property	Plain-language explanation
Durability	The ledger is copied across many nodes. Access depends on preserving valid keys.
Portability	A signed transaction can transfer value through a communications network.
Divisibility	One bitcoin divides into 100,000,000 sats.
Fungibility	Equal amounts are spendable under the same protocol rules, although public transaction history can affect how some services treat particular funds.
Scarcity	Issuance follows public rules with a 21 million maximum.
Verifiability	A node can verify supply, transaction rules, and ownership proofs.
Acceptability	Acceptance is growing in some places but remains much lower than national currencies in many communities.

Bitcoin can perform the three functions of money, but not equally in every place:

- As a **store of value**, its supply is predictable, but its market price is volatile.
- As a **medium of exchange**, it can move without a bank, but acceptance, fees, and payment tools vary.
- As a **unit of account**, amounts can be stated in bitcoin or sats, but most goods are still priced in fiat currency.

Bitcoin therefore presents both benefits and trade-offs.

Potential benefits include direct ownership, predictable supply, global transfer, open participation, transparent rules, and resistance to unilateral censorship.

Trade-offs include price volatility, personal security responsibility, irreversible errors, changing transaction fees, dependence on electricity and communications for digital transfer, public transaction records, and limited acceptance.

Whether you call Bitcoin sound money should follow from your criteria. Compare it with fiat money, precious metals, and other assets using the same standards.

Local-value check: Does bitcoin have value in your community? Separate three questions in your answer: What market price can be observed? Where, if any-

where, is it accepted? What personal preference or bias affects your judgment? Price, local usefulness, and personal approval are related but not identical.

2.14 Bitcoin, stablecoins, and other crypto-assets

Digital assets can look similar on a screen while relying on very different systems. Bitcoin has public supply rules and no issuer that can create new bitcoin outside those rules. A stablecoin normally depends on an issuer, reserves, legal claims, or a mechanism intended to track another currency. Other crypto-assets may have different people or organizations controlling issuance, changing rules, or operating important parts of the system.

When comparing any digital asset, ask: Who controls issuance or changes? What can I verify independently? Who holds the keys? Which company, issuer, or other counterparty must I trust? This Diploma uses the comparison only to clarify Bitcoin. It does not recommend acquiring, trading, or promoting any asset.

The examples below identify commonly used names and tickers only. They are not endorsements, recommendations, or a claim that any example is safe, reliable, or suitable for you.

Bitcoin (BTC) is a digital asset on the Bitcoin network. Its maximum supply and issuance schedule are public rules. People can hold bitcoin directly through keys and can independently verify Bitcoin rules by running suitable software. It can be sent globally without needing a central Bitcoin issuer to approve every transaction. Its market price can change sharply, and losing keys or recovery information can mean permanent loss; a mistaken payment is usually hard or impossible to reverse. Fees, confirmation time, electricity, communications, public transaction records, and local acceptance can also affect use. A wallet provider or exchange can add its own counterparty and custody risks.

Stablecoins, such as USD Coin (USDC) and Tether (USDT), are digital tokens designed to track a reference value, often one unit of a national currency such as the U.S. dollar. Some may be useful for moving a token value between compatible services or networks. Their design, reserves, redemption rights, and controls vary substantially. A stablecoin can lose its intended value ("depeg"), and users may depend on an issuer, reserve custodian, bank, legal redemption process, blockchain operator, or bridge. Issuers or platforms may freeze tokens or restrict access where their rules or laws allow. Reserve quality, transparency, insolvency, regulation, smart-contract errors, scams, and network fees are additional risks.

Other crypto-assets, such as Ether (ETH) and Solana (SOL), are a broad category of assets used by other networks or applications. Their rules may support functions such as paying network fees, staking, governance, or application use. Issuance, control, technical design, and what can be independently verified differ from one asset to another. Prices and liquidity can change rapidly, and developers, foundations, validators, administrators, or large holders may have significant influence, depending on the system. Rule changes, token issuance, smart-contract bugs, hacks, bridge failures, scams, platform failure, and unclear legal treatment can create loss or access risks. Holding a token does not automatically give a user ownership of a company, guaranteed income, or a right to redeem it for another asset.

Asset type	Similarity	Key difference
Bitcoin	A digital asset that can be held through keys.	Public supply rules; no central issuer.
Stablecoins	A digital token that can be held through keys.	Intended to track another value; relies on design and counterparties.
Other crypto-assets	Digital assets used on networks or applications.	Rules, issuance, and control vary by asset.

Before using any digital asset, distinguish the asset itself from the service holding it for you. A familiar ticker, a stable-looking price, or a well-designed app does not remove the need to identify and compare custody, counterparty, technical, legal, and loss risks.

2.15 Learning and responsible use are not trading or promotion

Learning how Bitcoin works is different from trying to profit from short-term price changes, persuading others to buy a token, or treating speculation as education. Trading and speculation can involve rapid decisions, loss, platform risk, and pressure to act without clear reasoning. Token promotion can create conflicts of interest and reward persuasion rather than clear explanation.

The purpose of this Diploma is to help you explain Bitcoin, ask better questions, and make safer decisions. It does not provide investment advice or tell you what to buy, sell, or hold.

2.16 Bitcoin origin timeline

Bitcoin appeared through a short series of public events.

Date	Event	Why it matters
31 October 2008	Satoshi Nakamoto published the whitepaper Bitcoin: A Peer-to-Peer Electronic Cash System.	The paper described direct online payments without a financial intermediary.
3 January 2009	Satoshi created the Genesis Block, the first block in Bitcoin's chain.	The Bitcoin network began. The block included a newspaper headline about bank bailouts, placing the launch in the context of the financial crisis.
12 January 2009	Satoshi sent bitcoin to Hal Finney in the first recorded person-to-person Bitcoin transaction.	The software moved value between two participants as designed.
22 May 2010	A user paid 10,000 bitcoin for two pizzas.	The event showed bitcoin being exchanged for ordinary goods and is remembered as Bitcoin Pizza Day.

Explore the source records: read the Bitcoin whitepaper, inspect the Genesis Block and the first recorded person-to-person transaction, and read the original pizza request. A block explorer is a convenient interface to public data, not a primary authority over Bitcoin's rules.

Satoshi Nakamoto is the name used by Bitcoin's creator or creators. Satoshi communicated online, released the software as open source, and later stopped participating. Satoshi's real-world identity remains unknown.

No central company owns Bitcoin. The software and rules continued because other people chose to run, review, and develop them. This origin matters: the creator did not remain as a permanent leader who could issue more coins or approve transactions.

The Genesis Block and the 2008 financial crisis give Bitcoin important historical context. However, evaluate Bitcoin by its design and operation, not only by a story about its creator.

At the end of this timeline is the event that made Bitcoin possible: in October 2008, Satoshi Nakamoto announced the Bitcoin whitepaper. It proposed a practical solution to a problem that had challenged digital money designers for decades: how to make online payments directly between people while preventing double spending without putting a central intermediary in control.

2.17 Participant empowerment, risk, and responsibility

Bitcoin's public rules, peer-to-peer network, and digital signatures give participants meaningful roles. Users can choose to receive, hold, and send bitcoin; nodes can independently verify the rules; miners can propose blocks; and developers can propose software changes. No single role controls the network for everyone.

These properties can empower participants, but they also place more responsibility on them. Without a central authority to approve transactions, recover access, or reverse [illegible] errors, each person must be able to identify the role and risks they accept.

For a person who chooses self-custody, control of the keys brings direct control of bitcoin. No company can freeze a correctly controlled on-chain balance. No company can restore a lost seed phrase either.

Self-custody is therefore not just a technical feature. It requires protecting backups, verifying addresses, checking wallet behaviour, and creating a recovery plan. The next module explains how to hold and use bitcoin carefully, including how to manage these risks and responsibilities.

Personal responsibility does not mean acting alone or refusing help. A cautious learner can use education, peer review, documented procedures, and small tests without giving anyone access to secret information.

2.18 Review: how does Bitcoin respond to fiat problems?

Pause and consider how Bitcoin changes the trust model: how do public rules, finite issuance, peer-to-peer transfer, and independent verification respond to the concerns raised in the previous module? Ask yourself how nodes and

miners have different roles, why double spending must be prevented, and which new responsibilities arise when a person controls their own keys. Bitcoin replaces some institutional trust with verification and personal responsibility; it does not remove risk or trade-offs.

Fiat-system issue	Bitcoin's response	New responsibility or trade-off
Supply can change through policy	Predictable issuance and a 21 million maximum	Users must verify and choose to enforce the rules
Payments depend on intermediaries	Peer-to-peer transaction network	Errors are usually irreversible
A central ledger keeper controls records	Independent nodes share and verify a public ledger	Public records can create privacy considerations
Access depends on an institution	Anyone can use the protocol and control keys	The user must secure keys and backups
A central party orders digital payments	Proof of work orders transactions without a central operator	Mining requires equipment and energy

Key thought: Based on what you have learned about Bitcoin in this module, does it have value to you? Does the current price of Bitcoin represent its value?

Key takeaways

- Bitcoin was created to make digital transfer possible without a central ledger keeper while preventing double spending.
- Its public ledger, digital signatures, peer-to-peer network, proof of work, and independent verification work together to maintain a shared transaction history.
- Bitcoin is decentralized in the sense that no single participant can unilaterally change balances or the rules for everyone.
- Users choose whether to hold or accept bitcoin; nodes verify the rules; miners propose blocks; and developers can propose, but not impose, software changes.

- Bitcoin has a public supply schedule with a maximum of 21 million bitcoin, and nodes reject blocks that break this rule.
- One bitcoin divides into 100,000,000 sats, so people can use smaller units without needing a whole bitcoin.
- Bitcoin can be evaluated as sound money using monetary properties, while still recognizing volatility, limited acceptance, irreversible errors, and personal responsibility.

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Module 03

Using Bitcoin Safely

3.1 Using Bitcoin Safely
3.2 Wallets
3.3 Addresses
3.4 Keys
3.5 Seed phrases
3.6 Custodial and non-custodial use
3.7 Hot and cold wallets
3.8 Wallet recovery and seed phrase testing
3.9 Receiving bitcoin
3.10 Sending bitcoin
3.11 Verification, block explorers, and [redacted] transactions
3.12 Safety checklist
3.13 Acquiring and exchanging bitcoin
3.14 On-chain and Lightning payments
3.15 Practical competency check observed by an educator
3.16 Review: what does self-custody responsibility mean?

3.1 Using Bitcoin Safely

This module is about safe first actions, not becoming an expert or taking financial risk. You will practise with a prepared simulation, test funds, or an amount approved by your educator. You ~~never~~ don't need to buy bitcoin to complete the Diploma.

What information can you share safely? What information could give another person control? How would you notice that a payment request has changed? These questions matter before any wallet, payment, or backup becomes important.

Use these three rules throughout the module:

1. **Share receiving information, never secrets.** An address or payment request can receive bitcoin. A private key or recovery phrase can control it.
2. **Pause before approving a send.** Check the person, payment method, destination, amount, and final screen.
3. **Keep mistakes small.** Stop and ask for help when a request is unclear, rushed, or asks for secret information.

By the end, you should be able to explain who controls the keys, complete a supported receive or send practice, respond safely to a scam or mismatch, and show recovery readiness without exposing any real secret. The goal is not speed. It is knowing when to pause.

The module follows three practical questions:

1. **What can I share, and what must remain secret?**
2. **What must I check before I receive or send?**
3. **When should I stop and ask for help?**

Use the explanations below to prepare for the three guided practices: a receive screen check, a send and verify check, and a recovery readiness or scam response scenario.

3.2 Wallets

A **Bitcoin wallet** is a tool that manages the information needed to receive and authorize Bitcoin transactions. It may be an app, a dedicated device, or a service. Bitcoin is recorded on the shared ledger; it is not stored inside a ~~phone~~ device.

What would happen if your phone stopped working? The Bitcoin record would still exist, but access would depend on the keys and recovery method the wallet uses. A wallet helps create receiving information, show balances and trans-

action history, and authorize spending. It does not remove the need to understand who controls access.

The first question about any wallet is: **who controls the keys?** Before using a tool, check who can authorize a payment, how access is recovered, and which payment methods it supports. Prefer well-established open-source wallet software where appropriate, while remembering that open source alone does not prove a tool is safe or suitable. Start with an educator's prepared environment; do not choose a tool because a stranger recommends it.

3.3 Addresses

A Bitcoin **address** is public receiving information for an on-chain payment. A wallet normally shows it as text and a QR code. It is not a password and cannot authorize spending by itself.

If you want to receive an on-chain payment, where should the address come from? Open the intended wallet and use its receive screen. A sender can scan the QR code or copy the address. This avoids the mistakes that can happen when long strings are typed by hand.

For every payment request, confirm the intended payment method. An on-chain address and a Lightning invoice are different. After scanning or copying, compare the destination with the trusted receiving screen before approving a payment. Why check after copying? Malicious software can replace copied address information. A new on-chain address for a new receipt can improve privacy. -> Maybe explain more?

When a payment matters, do not decide that an address is safe only because it looks familiar. Confirm the intended recipient, correct Bitcoin network, destination, and payment method. Use a small test first when the process or amount is unfamiliar.

3.4 Keys

Bitcoin ownership is enforced through cryptographic keys.

A **private key** is secret information that allows a wallet to create a valid digital signature. That signature authorizes the spending of bitcoin controlled by the corresponding key. Anyone who obtains the private key can usually spend those funds.

A **public key** is mathematically related to the private key. It helps verify a signature without revealing the private key. Addresses are created from public-key information.

The essential rule is simple:

Public receiving information can be shared. Private spending information must remain secret.

A digital signature proves that the required private key authorized a transaction. It does not place the private key inside the transaction. Nodes verify the signature before accepting the spend.

Modern wallets manage many keys automatically. Beginners should not copy or handle individual private key strings. Protect the wallet's approved backup and follow its recovery procedure.

No ~~legitimate~~ educator, wallet company, exchange employee, technical support worker, or Bitcoin developer needs your private key or seed phrase. Anyone who asks for it should be treated as a threat.

Try it: Sort prepared examples into three groups: safe to share for receiving, secret spending information, and information that may be public but should still be kept private when linked to a person.

3.5 Seed phrases

A **seed phrase**, also called a recovery phrase, is an ordered list of words that many wallets use to back up the information needed to recreate their keys. Common lengths are 12 or 24 words, although wallet designs differ.

The order and spelling of the words matter. If a person has the seed phrase, they can often restore the wallet on another compatible device and spend the funds. The seed phrase is therefore not only a backup. It is full access.

Protect a seed phrase with these rules:

- write it down privately and accurately during wallet creation;
- follow the wallet's official backup instructions;
- keep it offline;
- never photograph it, store it in cloud notes, email it, or send it in a message;
- never type it into a website or ~~ordinary~~ computer form;
- keep it out of cameras, microphones, other people's view, and places where it may be damaged, lost, or found.

Some wallets use additional passphrases, multiple keys, or other backup systems. These are Advanced learning. Do not add complexity you cannot explain and recover from safely. If a seed phrase may have been seen, copied, entered online, or stored on an untrusted device, treat it as compromised: seek guidance without revealing it and use a new trusted recovery process.

3.6 Custodial and non-custodial use

Custody answers a practical question: who controls the keys required to spend the bitcoin?

In a **custodial** service, a company or another person controls the keys. Your displayed balance is a promise from the custodian. You usually sign in with an account and ask the service to process a withdrawal or payment.

In a **non-custodial** or **self-custodial** wallet, you control the keys. The wallet signs transactions under your authority. There may still be software, hardware, server, or network dependencies, but no custodian must approve a valid (on-chain) spend.

Question	Custodial use	Self-custodial use
Who holds the keys?	The service	The user
Who handles recovery?	Usually the service through an account process	The user through a backup process
Main trade-off	Familiar support, with provider dependence	Direct control, with backup and error responsibility

Custody is not always obvious from the interface. If you cannot control the recovery method and need a company to approve a withdrawal, you are probably using a custodian. Self-custody is not a test of personal worth; the important step is to identify the trust and risk involved.

Try it: Read two prepared wallet descriptions. For each, state who controls the keys, who could restore access, and one risk to check before using it.

3.7 Hot and cold wallets

A **hot wallet** manages keys on a device connected to the internet, such as a phone or computer. It is convenient for regular payments. It also has a larger attack surface because malicious software, insecure updates, phishing, physical access, or device compromise may expose the wallet.

A **cold wallet** keeps the keys used for signing transactions away from an internet-connected general-purpose device. A **hardware wallet** is a dedicated device designed to generate, protect, and use keys while keeping them isolated from the connected computer or phone.

Arrangement	Common strength	Responsibility to remember
Hot wallet	Convenient access for regular payments	An internet-connected device can face malware, phishing, or loss.
Hardware or cold wallet	More isolated signing keys	The device, backup, and recovery process still need protection.
Custodial account	Familiar login and available support	A provider controls access and may impose limits

Try it: For a small, fictional payment and a less-frequent fictional payment, explain which trade-off matters more. Do not choose a tool or amount for your own funds.

3.8 Wallet recovery and seed phrase testing

What would you do if the original device failed? A recovery plan should let you restore the expected wallet without exposing the recovery information. Practice only with educator-prepared, empty, or simulated materials. Follow the wallet's official recovery method privately, confirm the expected training wallet, and reset or return the temporary material safely.

An educator can observe your process without seeing any recovery words. Never enter a real recovery phrase into a website, message, or ordinary computer form.

A safe recovery-readiness practice is simple:

1. Use the official instructions for the exact training wallet.
2. Confirm that the wallet and its recovery method are compatible.

3. Work privately, without cameras or observers seeing the secret material.
4. Use the wallet's approved backup check or recovery method.
5. Confirm the expected receiving information or training history.
6. Reset or clear the temporary training environment as instructed.

The point is not to make recovery feel easy. It is to understand the process before a device failure creates pressure.

3.9 Receiving bitcoin

Practice 1: receive safely

Receiving does not authorize funds to leave your wallet, but it still needs careful checking. Begin by asking: am I using the intended wallet and the intended payment method?

1. Open the prepared wallet and choose **Receive**.
2. Identify whether the request is on-chain or Lightning.
3. Share only the address, QR code, or payment request.
4. If an amount is shown, check it with the sender.
5. Check the amount and status in your own wallet.

A message or screenshot that says "sent" is not proof. What evidence can you see yourself? Check the receiving wallet or approved transaction view. For on-chain payments, a wallet may first show the transaction as pending. Confirmation status adds settlement confidence; it does not repair a wrong address or scam. A wallet may offer a fresh address for a new receipt; older addresses may still work, but fresh addresses can reduce easy links between payments.

explain.

3.10 Sending bitcoin

Practice 2: pause and check before sending

Sending needs more care because a valid Bitcoin transaction is usually irreversible. Before you touch the approve button, can you explain who will receive the payment and why?

1. Confirm the recipient and purpose through a trusted channel.
2. Confirm the payment method: on-chain or Lightning.
3. Scan or enter the destination, then compare it with the trusted receiving screen.
4. Check the amount, unit, fee or expiry information, and final summary.
5. Use a small test when the process is unfamiliar.

6. Approve only when every item matches; then check the payment result.

For an on-chain payment, the **network fee** is paid for including the transaction in a block. It depends mainly on the transaction data and demand for block space, not simply on the amount being sent. Fee estimates and timing can change, so read the final summary rather than assuming that a low-value payment has a low fee.

If the destination changes unexpectedly, the request becomes urgent, or anyone asks for recovery information, what should you do? Stop. Verify the request independently, or ask for help without sharing secrets.

3.11 Verification, block explorers, and irreversible transactions

A **transaction ID (TXID)** identifies an on-chain transaction. A **block explorer** shows public Bitcoin data, such as a transaction's status and confirmation count. It is a useful view, not an authority over Bitcoin and not proof of the recipient's identity or honesty.

What can this public information show, and what can it not show? It can show a transaction's status, destination, amount, fee, and confirmation count; it cannot prove why a payment happened or whether a person is trustworthy. An on-chain transaction is unconfirmed after it is broadcast and gains its first confirmation when included in a valid block. Later blocks add confidence to its place in the record, but they do not fix a wrong destination or a scam. There is no universal confirmation count for every payment; the suitable check depends on the amount and risk.

To inspect a prepared transaction, obtain its TXID, open a trusted explorer for the intended Bitcoin network, and check the status, destination, amount, fee, and confirmations. An explorer is only an interface. Searching an address or TXID may reveal to its operator that your device is interested in that information. For practice, use an educator-prepared transaction or printed screenshot. Do not search a learner's personal address or transaction on a public service.

3.12 Safety checklist

Practice 3: spot a risk and stop safely

What would make you stop before a payment or installation? Treat these as red flags: someone impersonates support or an educator; creates urgency or secrecy; promises guaranteed returns; asks you to send bitcoin first; changes a

destination; or asks for keys, recovery words, remote access, or an advance fee.

The safe response is always: **Pause. Share nothing. Send nothing. Close the channel. Verify independently through a known source. Ask for help without revealing secrets.** A real opportunity can survive a careful check; a scam often depends on preventing one.

Protect secrets by keeping seed phrases and private keys offline and out of view. Verify destinations by checking the payment method, recipient, address or invoice, amount, and final screen after copying or scanning. Use official instructions and a verified source for any wallet or service; be cautious with sponsored search results, direct-message links, and look-alike sites. If the process differs from the demonstrated process, pause rather than trying to solve it under pressure. Start with safe practice, a simulation, or a small approved amount; never borrow money to experiment.

Try it: In a prepared message or payment screen, identify the warning sign, what could be lost, and the first safe action.

3.13 Acquiring and exchanging bitcoin

Pathway awareness, not financial advice

People may receive bitcoin through work, a community payment, a peer exchange, an ATM, or a service. Each pathway has different costs, custody, privacy, fraud, legal, and operational trade-offs. This Diploma does not recommend a provider or tell you which path to use.

- **Earning or receiving:** Bitcoin may be received for work, goods, services, a gift, a refund, or a community payment. The parties still need to agree on the amount, timing, and payment method.
- **Peer-to-peer exchange:** Two people may exchange fiat money or goods for bitcoin directly or through a matching service. This can introduce counterparty, physical-safety, payment-reversal, and fraud risks.
- **Exchange, broker, ATM, or kiosk:** A service or machine may provide a familiar route, but may involve fees, identity checks, custody, withdrawal limits, or a poor exchange rate.

Before comparing a path, ask:

Risk area	Question to ask
Costs	What price, spread, and service or withdrawal fees apply?
Custody	Who controls the bitcoin before and after the exchange?
Identity and privacy	What personal information is collected, stored, or shared?
Withdrawal and settlement	Can payment or withdrawal be delayed, limited, or reversed?
Counterparty and safety	How can I verify the other party or service, and is an in-person exchange safe?
Local context	Is the route lawful and appropriate where I live, and what records may matter?

No pathway is automatically best everywhere. Local examples and provider-specific guidance belong in Module 4. **Know your customer (KYC)** describes identity checks a regulated service may require. It can support account recovery or legal compliance, while also creating a record that connects identity and financial activity. Paths with less identity collection can involve different counterparty, legal, safety, cost, or complexity risks.

3.14 On-chain and Lightning payments

Bitcoin has more than one payment path. **On-chain** payments are recorded in Bitcoin's blockchain and gain confirmations. **Lightning** is a payment network built on Bitcoin that can support fast, frequent payments. A Lightning payment may complete quickly or fail, and its wallet can still be custodial or self-custodial.

Question	On-chain Bitcoin	Lightning
Receiving information	Bitcoin address or on-chain payment request	Lightning invoice, offer, or supported Lightning address
Payment result	Broadcast first, then confirmations	Usually completes or fails quickly
Common use	Base-layer settlement or less frequent transfer	Fast, smaller, or frequent payment
Beginner checks	Correct address, network, amount, fee, and confirmation status	Correct invoice or request, amount, expiry, custody, and payment result

Which path does the recipient expect? Confirm whether they have provided an on-chain address or a Lightning invoice or request before paying. An on-chain wallet cannot directly pay every Lightning invoice, and a Lightning wallet may not support every on-chain action. Some applications support both paths but may add fees or service dependencies. Detailed Lightning setup, channels, routing, and liquidity are Advanced learning; live local practice belongs in Module 4 when it is safe and appropriate.

Imagine that a prepared scenario offers both payment paths. Which request can the intended wallet use, and what would you check before approving it? The answer depends on the receiving request, the wallet, and the situation—not on a universal "best" path.

3.15 Practical competency check observed by an educator

Before assessment, you will receive supported practice and know the task and safety boundaries. Your educator—not an app, attendance record, or self-report—will observe you using an approved prepared, simulated, or low-stakes environment.

You will show that you can open or navigate the prepared wallet, complete an approved first-access pathway, locate and check receiving information, and receive bitcoin or safely simulate receipt. You will also complete or safely simulate a receive or send flow and show recovery readiness with prepared materials while keeping all secrets private. You may practise responding safely to a mismatch or scam through guided scenarios. If you are not yet ready, you

should receive feedback, more practice, and another opportunity to demonstrate the skill.

3.16 Review: what does self-custody responsibility mean?

Imagine that a friend sends you a payment request while an unfamiliar support account says you must act immediately. What could you check before doing anything? Can you protect a recovery plan, distinguish receiving information from secrets, and verify the person, payment method, destination, amount, and final screen before sending? Whether a service or you control the keys, the practical recap is the same: protect secrets, verify before approving, recognize scam signals, and keep mistakes small while learning.

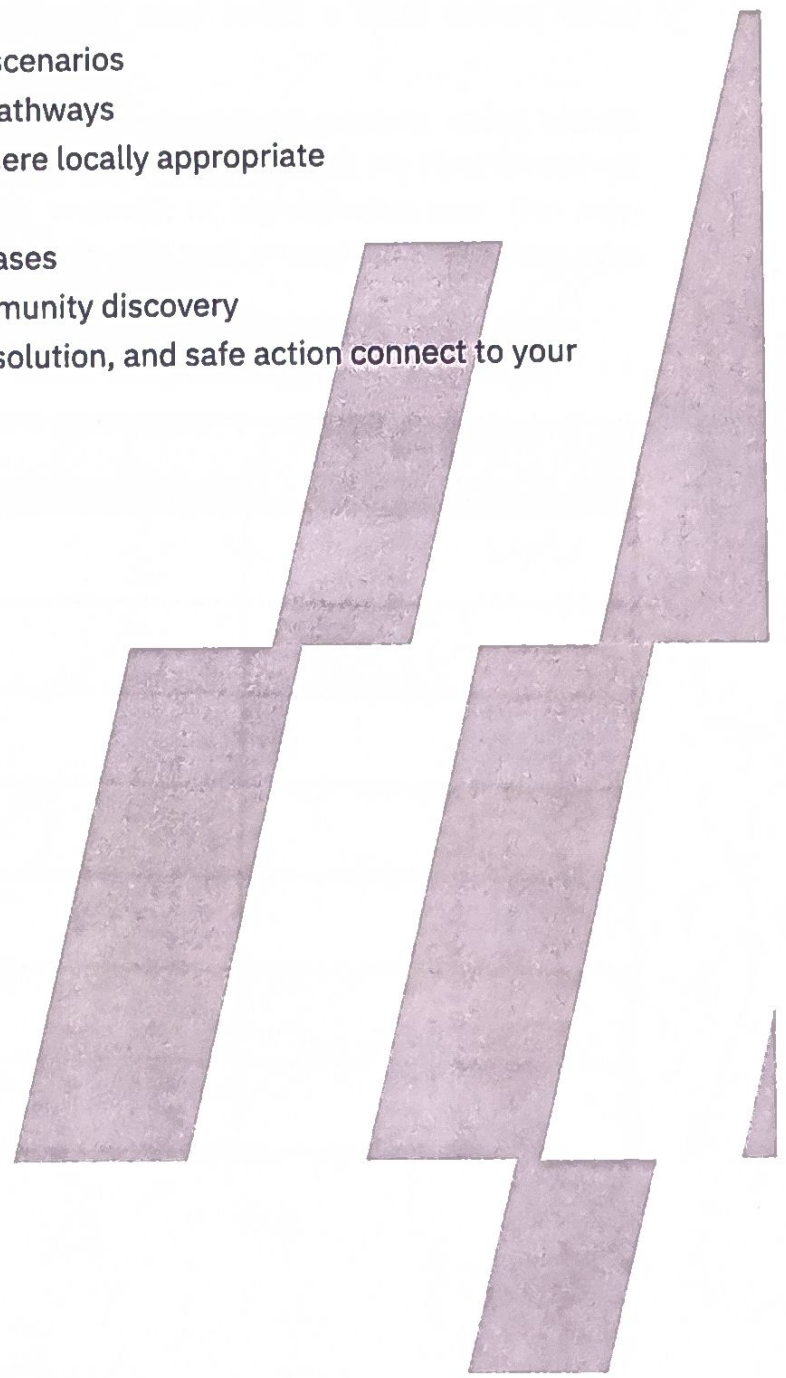
Key takeaways

- A wallet manages the keys and information used to receive, check, and spend bitcoin; it does not store bitcoin inside a phone or device.
- Addresses and payment requests can be shared to receive bitcoin, but private keys and seed phrases must remain secret.
- A seed phrase can provide full recovery access, so it must stay offline, private, and out of websites and messages.
- Custodial and self-custodial arrangements create different recovery and control responsibilities. Hardware wallets still depend on a safe backup.
- Before receiving or sending, check the correct payment method, destination, amount, and final status; a valid send is usually irreversible.
- Start with safe practice or small approved amounts. Pause when a request creates urgency, secrecy, or pressure.

Module 04

Bitcoin in Your Community

4.1 Bitcoin in Your Community
4.2 Your local Module 4
4.3 Practical workshops and demonstrations
4.4 Local Bitcoin tools and resources
4.5 Lightning payments where locally appropriate
4.6 Daily-life Bitcoin use
4.7 Local spending and receiving scenarios
4.8 Local on-ramp and off-ramp pathways
4.9 KYC and no-KYC trade-offs where locally appropriate
4.10 Circular economies
4.11 Merchant or community use cases
4.12 Local resource maps and community discovery
4.13 Review: how do the problem, solution, and safe action connect to your community?



4.1 Bitcoin in Your Community

Learner promise: You will see how Bitcoin can function as money in your local context.

This is the **Connect** stage of the Diploma. You will connect the money problem, Bitcoin response, and safe use responsibility from Modules 1–3 to a local need or opportunity, a realistic alternative, and a safe practical task. You are not expected to adopt Bitcoin or conclude that it is the best option in every local situation.

Bitcoin is one global protocol, but people use it under different local conditions. Internet access, phones, laws, taxes, currencies, banks, merchants, fees, language, safety, and technical support all affect what is useful.

Your educator selects the learning activities in this module. The goal is not to demonstrate every tool. It is to connect concepts from Modules 1–3 to real choices in your community. The educator may invite a local expert when specialized knowledge is needed.

Use the Module 3 safety checklist throughout every local module. Using bitcoin or a named tool in a supervised exercise does not mean that My First Bitcoin or your educator recommends frequent, ongoing, or higher-value use. The purpose is to compare the option and its trade-offs well enough to make your own decision.

Record the focus of your Module 4:

Local learning choice	Your notes
Module or workshop	
Local problem or opportunity	
Tool, resource, or scenario	
Main benefit to examine	
Main risk to examine	
Evidence or result	
Safe next step	

improve?

4.2 Your local Module 4

In this module, you will apply an idea from Modules 1–3 to a real need or opportunity where you live. You may explore earning, saving, receiving, spending, merchant use, community resources, or another local use of Bitcoin as money.

This matters because Bitcoin is experienced locally. You will better understand what is useful, what is not, and which conditions and risks shape a responsible choice. You will also become familiar with local Bitcoin tools, resources, and organisations without being expected to use or recommend them.

Your educator will guide you because they understand the local context. They may invite a local expert to help you learn from experience and interact positively and respectfully with your community. The goal is to build informed connections and help foster community, not simply to complete a transaction.

Your local Module 4 activity will help you:

1. **Connect:** Identify the money problem, Bitcoin response, and safe action that matter locally.
2. **Compare:** Consider local conditions and compare Bitcoin with a realistic existing option.
3. **Practise safely:** Observe, compare, or try a defined task with guidance.
4. **Explain:** Identify risks, show what you learned, and name a responsible next step, if any.

Your activity may explore questions such as:

- Can a local worker receive bitcoin for a service?
- Can a family compare a remittance path with an existing local method?
- Can a merchant accept payment without holding a large bitcoin balance?
- Can a community group receive support while keeping custody and records clear?
- Can you identify reliable Bitcoin support in the community?
- Can saving in sats address a local problem, and what new risk does it create?

A local example is not proof that the same method works everywhere. Explain what is specific to your community.

4.3 Practical workshops and demonstrations

You will move through three stages: **Presentation**, where your educator shows the process and its risks; **Practice**, where you follow a guided process using

safe materials or a very small amount; and **Production**, where you complete a task or explain a decision independently.

Your workshop may include:

- identifying whether a wallet is custodial or self-custodial;
- locating the receive screen without creating or exposing a real seed phrase;
- comparing an on-chain address with a Lightning invoice;
- checking a transaction in a wallet or block explorer;
- comparing fees and received amounts across local pathways;
- identifying scam signals;
- planning a merchant payment or community donation flow;
- mapping local resources or payment relationships.

Before any live demonstration, make sure you know:

- the tool and instructions were checked recently;
- how the educator will explain recovery and what happens if something fails;
- whether real funds are involved;
- that you will not be pressured to install software or disclose financial data;
- that any amount is small enough for a complete loss to remain a lesson, not a crisis;
- that an offline or no-funds alternative is available;
- how the class will stop if the result differs from the plan.

The result of your workshop should be an explanation or skill, not only a completed transaction.

4.4 Local Bitcoin tools and resources

Local bitcoiners may use wallets, exchanges, peer-to-peer markets, merchant tools, maps, meetups, hardware devices, and education resources. Availability and quality can change quickly.

Use this comparison before adopting a tool:

Area	What to verify
Purpose	What problem does the tool solve? Is it necessary?
Custody	Who controls keys and funds?
Source	Who publishes it? Is the domain or application listing genuine?
Code	Is it open source? Is it actively reviewed and maintained?
Recovery	What happens if the device, service, or company disappears?
Privacy	What identity, transaction, contact, or device data are collected?
Cost	What fees, exchange spreads, subscriptions, or hardware costs apply?
Compatibility	Does it support the required wallet, network, address, or invoice?
Support	Is safe local help available without sharing secrets?
Law	Are there local restrictions, reporting duties, or tax effects?

Named tools used in class are local examples, not endorsements by My First Bitcoin. Record the date when a tool was checked. A tool that was suitable last year may have different ownership, fees, features, or risks today.

Local tool record

Question	Finding
Tool and version	
Date checked	
Custody model	
Main use	
Main benefit	
Main risk	
Recovery method	
Alternative	

4.5 Lightning payments where ~~locally~~ appropriate

Lightning can make fast, small Bitcoin payments practical. It may be suitable for a merchant counter, repeated local payments, online services, or transfers where an on-chain fee would be too large in relation to the payment.

A local Lightning demonstration should identify:

- whether the wallet is custodial or self-custodial;
- how the wallet is funded;
- whether funding requires an on-chain transaction or a service;
- what backup or recovery protects the balance;
- what type of payment request is being used;
- what fees or limits apply;
- what happens when a payment fails;
- how the recipient confirms payment.

A **Lightning invoice** is a payment request created for a specific payment. It can include an amount, destination information, and an expiry time. Read the wallet's confirmation screen before paying.

A **Lightning address** is a human-readable identifier supported by some services. It may look like an email address, but it is a payment-routing convenience, not an email account. It depends on the provider or domain that supports it.

For a guided payment:

1. The receiver creates the correct request.
2. The sender scans or copies it.
3. The sender checks the recipient, amount, unit, and fee.
4. The sender approves only after the details are correct.
5. Both sides confirm the result in their own wallets.
6. The group explains the custody and failure risks.

Do not assume that a successful Lightning payment proves the wallet is safe for larger balances. Payment convenience and custody security are separate questions.

4.6 Daily-life Bitcoin use

Bitcoin can function as money in several ways. Spending is only one of them.

Earning: A worker, creator, business, or community member receives bitcoin for value provided. Earning can avoid a separate purchase process, but the receiver must agree on pricing, volatility, records, and custody.

Saving: A person holds bitcoin for future use. Predictable supply may be attractive, but the fiat exchange value can fall sharply and key security becomes important.

Receiving: A person accepts a gift, donation, refund, remittance, or repayment. The receiver must provide the correct payment request and verify settlement.

Sending: A person transfers value to family, friends, a business, or an organization. They must compare payment path, speed, fee, privacy, and finality.

Spending: A person exchanges bitcoin for goods or services. They should identify the exchange rate, merchant policy, payment method, and local tax consequences that apply.

Pricing and accounting: A person or business records value in bitcoin or sats. In many places, the fiat value at the time of payment is still needed for accounts or taxes.

Donating: A person supports an organization or individual directly. They should verify the recipient's payment request through a trusted source.

For any use, ask three questions:

1. What existing problem does Bitcoin address here?
2. Is Bitcoin better than the available alternative for this case?
3. What new responsibility or risk does Bitcoin create?

Avoid starting with the tool. Start with the need.

4.7 Local spending and receiving scenarios

Use locally familiar goods and services. The exact item does not matter; the payment reasoning does.

Scenario 1: A small purchase

A customer wants to buy an everyday item. The merchant accepts Lightning and on-chain Bitcoin.

- Which path is likely to suit a small, immediate payment?
- Who creates the payment request?
- How does the customer check the amount and unit?
- How does the merchant know the payment succeeded?
- What happens if the customer's wallet shows “sent” but the merchant does not see “received”?

Scenario 2: Payment for work

A worker agrees to receive part of a payment in bitcoin.

- Is the price fixed in local currency, bitcoin, or sats?
- At what time is the exchange rate determined?
- Who pays any fee?
- Is the receiver using self-custody?
- What income or tax record is required?
- Can the worker safely manage the amount?

Scenario 3: A transfer between family members

One person wants to send value to a family member in another place.

- What are the costs and delays of the current method?
- How will each person obtain or use bitcoin?
- Does the recipient want bitcoin or local currency?
- Which exchange and custody risks appear at each step?
- Does the full route save time or money after all fees are included?

Scenario 4: A refund

A merchant needs to return value after an on-chain payment.

- Why should the merchant request a fresh refund address?
- How can the merchant verify that the request really came from the customer?
- Who pays the return transaction fee?
- What record should both parties keep?

Scenario 5: A community donation

A local organization wants to receive support in bitcoin.

- How will donors verify the real payment request?
- Who controls the receiving wallet and backup?
- Will donations be received on-chain, through Lightning, or both?
- What information should be public for transparency, and what should remain private for donor safety?
- What records are needed for local accounting, tax, or reporting rules?

For a live practice, use a low-value item or a simulation. The learner should be able to explain the complete flow before approving payment.

4.8 Local on-ramp and off-ramp pathways

An **on-ramp** is a path from local money, work, or another asset into bitcoin. An **off-ramp** is a path from bitcoin into local money, goods, or services. The same pathway may perform both roles.

Local pathways may include:

- earning bitcoin;
- buying or selling directly with another person;
- using an exchange or broker;
- using an ATM or kiosk;
- accepting or spending with a merchant;
- using a payment service that converts between bitcoin and fiat;
- participating in a local circular economy.

Compare the **complete route**, not only the first fee. A service may advertise a low purchase fee but use a poor exchange rate or charge for withdrawal. A peer-to-peer path may appear private but require time, trust, and physical safety.

Pathway	Total received	Time	Custody	Privacy	Reversal risk	Main failure point
Local option A						
Local option B						
Existing non-Bitcoin option						

Check whether the path allows withdrawal to a wallet you control. A displayed bitcoin balance inside a closed service may be an internal claim rather than an on-chain balance under your keys.

For in-person exchange, use a public and appropriate location, avoid carrying unnecessary cash, bring a trusted person where suitable, verify payment before leaving, and follow local law. Do not reveal where you store bitcoin or how much you own.

4.9 KYC and no-KYC trade-offs where locally appropriate

KYC and no-KYC pathways make different trade-offs. Neither label removes the need for careful evaluation.

A KYC service may request a legal name, identity document, address, photograph, source-of-funds information, or other records. This may allow the service to meet local rules, connect to banks, set higher limits, or support account recovery.

KYC also creates data risk. The service may store a permanent connection among identity, payment methods, and bitcoin transactions. A breach, legal request, internal misuse, or change in policy may expose or restrict that information.

A no-KYC pathway may collect less identity information. This can improve financial privacy. It may also involve less regulated counterparties, lower liquidity, higher prices, reduced support, greater fraud risk, or legal restrictions.

Compare the real path:

Question	KYC path	No-KYC path
What identity data are collected?		
Who stores the data and for how long?		
Can the transaction be linked on-chain?		
What fraud protection exists?		
What counterparty risk exists?		
What limits and fees apply?		
What local law applies?		
What happens during a dispute?		

Privacy is not secrecy from every risk. It is the ability to control information about yourself. Do not break local law or expose yourself to physical danger in pursuit of privacy. Seek current local guidance from a qualified person when the rules are unclear.

4.10 Circular economies

A **Bitcoin circular economy** is a network in which people earn, spend, save, and exchange bitcoin within a community. Value circulates among workers, customers, merchants, producers, educators, and service providers instead of entering only through a one-time purchase and leaving through immediate conversion.

A circular economy can reduce dependence on distant financial services and create direct local relationships. It can also fail if there is little demand, poor education, unreliable tools, weak security, or pressure to use bitcoin without a real need.

Map a simple local cycle:

1. Who can earn bitcoin?
2. Where can they spend or use it?
3. Which merchants can pay a supplier or worker in bitcoin?

4. Which participants need to exchange to or from fiat?
5. Who provides education and technical support?
6. What happens when internet access fails?
7. What custody and price risks does each participant carry?

A healthy circular economy depends on voluntary participation. Merchants and workers should be able to identify the payment, pricing, custody, accounting, and volatility risks. Adoption numbers alone do not show whether the network is useful.

The strongest local loop solves a real problem for more than one participant. It should not depend permanently on giveaways or on one organizer processing every payment.

4.11 Merchant or community use cases

A merchant needs more than a receive button. The full payment process may include pricing, staff training, receipts, refunds, custody, conversion, accounting, taxes, device security, and support.

Before accepting Bitcoin, a merchant should decide:

- whether prices are displayed in local currency, sats, or both;
- whether payments use on-chain Bitcoin, Lightning, or both;
- whether the wallet is custodial or self-custodial;
- who can access the receiving device;
- how received funds are secured or moved;
- whether some funds are converted to local currency;
- how payment is confirmed at the point of sale;
- how refunds and mistakes are handled;
- what records local law requires;
- what staff do when the network or wallet fails.

Community use cases may include:

- paying local workers or creators;
- receiving donations transparently;
- transferring value between community groups;
- supporting a savings group with clearly defined custody;
- helping a remote worker receive international payment;
- connecting local producers and merchants;
- teaching safe wallet use;

- providing emergency support where ordinary payment systems are unavailable

Community custody requires special care. If one organizer holds funds for many people, that organizer becomes a custodian and a point of trust. Shared funds need transparent rules, records, access controls, and legal review where appropriate.

For each use case, define success in terms of the problem solved, not the number of transactions completed.

4.12 Local resource maps and community discovery

Maps and community directories can help learners find merchants, educators, meetups, and circular economies. Listings can be incomplete or outdated. A place shown on a map may have stopped accepting Bitcoin, changed its wallet, or closed.

Use a resource map safely:

1. Check when the listing was last verified.
2. Contact the merchant or organizer before making a special journey.
3. Ask which payment method is accepted.
4. Bring an alternative way to pay.
5. Do not publish another person's location or details without permission.
6. Submit corrections only when the information has been checked.

A local community can provide practical support, but membership is not proof of trustworthiness. Do not reveal seed phrases, balances, home addresses, or security arrangements at a meetup. Be cautious when a new contact offers investment management, private exchange, guaranteed returns, or urgent help.

Record your local resources:

Resource	Purpose	Date verified	Contact or location	Safety note
Education group				
Merchant directory				
Technical support				
Local exchange information				
Legal or tax information				

Good community discovery helps you find people who explain, verify, and respect your choices. It should not make you dependent on one leader.

4.13 Review: how do the problem, solution, and safe action connect to your community?

Complete this review individually, then compare your reasoning with your team or cohort. Consider the local example you studied: what real need did it address, which function of money was involved, and was value moved on-chain, through Lightning, or both? Ask who controlled the keys, what the strongest benefit and most serious risk were, and how local law, infrastructure, language, or culture affected the result. A responsible next step is one that fits the local context, makes costs and custody clear, limits errors and scams, and can be evaluated against evidence.

Module 4 answer: Bitcoin can function as money when people voluntarily use it to earn, save, receive, send, spend, price, or settle value. The Connect stage begins with a real need, compares a realistic alternative, makes custody and costs clear, respects local law and safety, and gives learners enough knowledge to choose freely.

Key takeaways

- Bitcoin as money is local: the useful pathway depends on community needs, infrastructure, laws, costs, tools, language, and safety conditions.

- An educator-curated Module 4 Connect activity should link the Problem, Solution, and Action learning to a real problem or opportunity, realistic alternative, safe practice, and evidence of learning.
- Local workshops should use a safe sequence: observe the process, practise with support, then explain or complete a defined task without exposing secrets.
- Before using a local tool or service, check its purpose, custody model, source, recovery process, privacy, costs, compatibility, support, and current local rules.
- On-chain and Lightning payments make different trade-offs in settlement, speed, fees, custody, and failure points; confirm which payment method a recipient expects.
- Local exchange pathways, KYC choices, merchant flows, and community use cases should be compared through their full costs, custody, privacy, safety, and legal trade-offs.
- A strong circular economy or community use case is voluntary, solves a real problem, protects participants, and does not make learners dependent on one service, organizer, or promise.

Appendices

Next Steps

Next Steps

Review your Learner Journey

You now have a foundation. Protect it by moving slowly.

First, return to the answers you wrote before the course. Do not erase them. Record your current answer and the evidence or idea that most influenced any change.

Opening question	My answer now	What changed or became clearer?
What is money?		
Why do people need it?		
What makes something useful as money?		
Who can create or control the money I use?		
What makes people accept or value it?		
Has its purchasing power changed?		

A Next Steps Checklist

☐ **Review and research.** Revisit your opening answers and identify what you can now explain, compare, or do about money, control, value, and trust. Check one question against a reliable source.

☐ **Practise or use one skill safely.** Repeat a concrete task such as identifying custody, checking an address, reading a transaction, recognizing a scam pattern, or comparing local pathways.

☐ **Prepare before taking on more risk.** Create a personal safety plan and learn more about self-custody, recovery, privacy, and inheritance before meaningful funds depend on your decisions.

☐ **Learn more.** Choose an advanced topic because it supports a real goal, not because online attention makes it feel urgent.
☐ **Connect with a responsible community.** Look for educators who welcome questions, explain trade-offs, and never ask for secret information. You can also explore the My First Bitcoin Education Network.
☐ **Explain Bitcoin to another person.** Use plain language and invite questions. A clear beginner explanation names the problem Bitcoin addresses, describes the network's basic response, and states an important trade-off or responsibility without hiding behind jargon.
☐ **Contribute or become an educator.** Improve the Diploma, translate or proofread a resource, share feedback, support a learning community, or explore the Educator Training Program.

} Links

One safe next step is enough. You do not need to perform every possible Bitcoin action immediately.

Name or team: _____ **Date:** _____

My chosen next step: _____

Evidence that I completed it: _____

Contribute to My First Bitcoin ~~and the Resources Hub~~

Choose a way to help and use the official forms to get started:

- **Proofread:** report unclear wording, errors, or translation improvements.
- **Volunteer:** offer time or skills to support the Diploma, educators, or local learning communities.
- **Share feedback:** tell My First Bitcoin what worked, what was missing, or what could be safer or clearer.
- **Contribute to the Resources Hub:** share a useful Bitcoin education resource, local learning resource, or improvement that other learners and educators can review and use.
- **Explore the Educator Training Program (ETP):** learn how My First Bitcoin prepares people to teach the Bitcoin Diploma.

- Proofread, volunteer, contribute to the Resources Hub, or share feedback
- Explore the Educator Training Program

Use the official forms only. Do not include seed phrases, private keys, wallet balances, or other sensitive personal or security information in a submission.

Continue with the Education Network

The My First Bitcoin Education Network connects independent educators and local learning communities. Explore it to learn how communities share experience, adapt open-source materials responsibly, and support Bitcoin education in their own context.

If you want to teach, start by building a strong foundation as a learner, then review the Educator Training Program (ETP). The ETP is a training path for prospective independent educators; participation, timing, and local requirements may change, so check the current official information before applying.

Learn from community case studies

Real-world stories can show how different communities have used the Diploma, but they are examples, not promises of the same result elsewhere. Read the current community case studies and consider the local need, evidence, trade-offs, and safety practices behind each one.

New case studies can help other communities learn. If you have a story or evidence that could become a responsible case study, submit it through the official form. Do not include sensitive learner, financial, or security information.

The My First Bitcoin Leaderboard is a separate public destination for current community activities, recognition, or possible perks. Any available perks, eligibility rules, and participation requirements are determined there and may change. They are optional and do not affect Diploma completion, assessment, certification, or access to the Education Network.

A safety reminder

Remember these rules:

- A seed phrase or private key gives control of funds. Never share it.
- Verify the recipient, network, address or invoice, amount, and fee before sending.
- Bitcoin transactions are usually irreversible.
- Use a small test when the process is new.
- Do not act under pressure, fear, or promises of guaranteed profit.
- Verify wallet software and support contacts through independent sources.
- Identify who holds the keys.
- Test recovery safely before meaningful funds depend on it.
- Keep financial balances and security arrangements private.

- Laws, taxes, fees, and services change. Check current local information.

Education reduces risk but cannot remove it. If a situation is unclear, stop and learn more before acting.

Connect with My First Bitcoin

My First Bitcoin provides open-source learning resources and supports independent educators and communities.

- Website: myfirstbitcoin.org
- Bitcoin Diploma page: ~~programs.~~myfirstbitcoin.org/programs/bitcoin-diploma
- Open-source organization: github.com/MyFirstBitcoin
- Curriculum source: github.com/MyFirstBitcoin/mfb-curriculum

Use the official website to find current programs, educator information, community links, and contact details. Verify the domain before entering personal information or following a donation request.

Source or update page, if included

The Bitcoin Diploma is a living open-source curriculum. Corrections and improvements can be reviewed publicly.

The canonical curriculum source is:

github.com/MyFirstBitcoin/mfb-curriculum

Check the edition or release information before teaching, translating, or adapting the material. If you find an error, use the repository's current contribution or issue process.

When adapting this work:

- preserve the required Diploma competencies and safety rules;
- identify local examples clearly;
- verify current claims, services, laws, and fees;
- use consistent beginner terminology;
- state what you changed;
- attribute My First Bitcoin;
- publish the adaptation under the required open-source license.

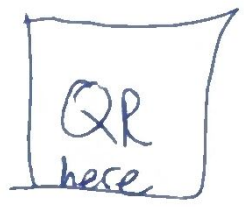
Optional donate / support QR

My First Bitcoin's educational resources are free and open source. Voluntary support helps maintain the curriculum, train educators, and expand access.

Use the **Support Us** section of the official website if you choose to contribute:

myfirstbitcoin.org

A printed edition may place one official support QR code here. Before paying, confirm that the QR destination uses the official domain or an official payment page linked from it. Check the recipient and amount in your wallet. A donation is optional and is not required for course completion or certification.



Certificate verification

A certificate is issued only after you meet the approved theoretical and practical requirements through the Diploma's educator-led assessment process. Optional app use does not affect certification. The certificate represents assessed beginner competence under supported conditions, not investment skill, professional financial competence, or readiness to manage high-value self-custody.

If your Diploma certificate includes a verification code, link, QR code, digital signature, or timestamp, follow the instructions printed on that certificate. Verify that the destination belongs to My First Bitcoin or the clearly identified issuing organization.

When checking a certificate through the official verification process, confirm:

- the learner name or identifier;
- the course and edition;
- the issuing organization and, where shown, the educator;
- the completion date;
- the verification status;
- whether the record has been changed or revoked.

Do not publish unnecessary personal information during verification. If no verification method is provided, contact the issuing educator through an independently verified channel.

Your most important result is not the certificate. It is the ability to explain money and Bitcoin clearly, recognize risk, verify before acting, and continue learning independently.